Internal Hackathon: Traditional ML Division

Deadline: 30 September, 11:59 PM

Last Updated: 28th September, 08:10 PM *(periodically check this!)*

Problem Description

Develop an ML-based solution to recommend an **Optimal Fantasy Team** of 11 cricket players for upcoming matches. Your solution must include a model pipeline that takes match details and the two competing squads as input and predicts the optimal fantasy team composition.

Key Constraints & Rules

- **Solo Participation:** This is strictly an individual hackathon.
- **Training Data Cutoff:** No data after **2025-12-31** may be used for training. This cutoff applies to all data sources used, including any external ones.
- **Pre-Toss Assumption:** Assume the toss has not happened. You will not have the final playing XI, so you must select from the full squad of 15+ members.
- **Data Sources:** Match data must come exclusively from cricsheet.org. You may supplement this with external public data (e.g., weather, pitch reports), provided it is publicly available, and **respects the same training cutoff** (no feature may encode information that would not have been knowable before the match date it's attached to).
- Participants may only use **all T20 and T20I matches** available on cricsheet dated on or before 31 December 2025.

Team Composition Rules

Your model's final output must adhere to standard fantasy composition rules, producing a team of exactly 11 players:

- **Wicket-Keeper:** 1–4 players
- **Batsmen:** 3–6 players
- **All-Rounders:** 1–4 players
- **Bowlers:** 3–6 players
- At least 1 player must be selected from each of the two competing teams.

Evaluation Output & The Hidden Test Set

Your code should programmatically train the model on data up to 2025-12-31. To ensure models generalise well and to prevent data leakage, we are implementing a blind evaluation phase.

Hidden Evaluation Dataset — released after 30th September

A hidden test.json file will be released to all participants on this date. Your evaluation pipeline must ingest this file and output your final prediction CSV before the submission deadline.

Your evaluation pipeline must ingest `test_fixtures.json` and programmatically generate a CSV containing:

- Match Date, Team 1, Team 2
- Predicted Best 11 Players (in descending order of predicted fantasy points), **with each player's role**
- Predicted Points of each player

Note for internal validation: For your own testing and report-building, your pipeline should also compute MAE against known historical matches, plus at least one alternative metric (e.g., RMSE, Rank Correlation). Briefly justify in your report which metric is more suitable for evaluating fantasy sports predictions.

Point system rules:

Each player's fantasy score will be calculated using the following scoring system. The final score of a selected fantasy team is the sum of the fantasy points of its 11 selected players. No Captain or Vice-Captain multipliers are used.

Batting: Run +1; Boundary Bonus +1; Six Bonus +2; 25 Run Bonus +5; 50 Run Bonus +10; 75 Run Bonus +15; 100 Run Bonus +20; Duck −2.

Strike Rate: Minimum 10 balls faced. >170: +5; 150.01–170: +3; 130–150: +1; 70.01–129.99: 0; 60–70: −1; 50–59.99: −3; <50: −5.

Bowling: Dot Ball +1; Wicket (excluding Run Out) +25; LBW/Bowled Bonus +5; 3-Wicket Bonus +5; 4-Wicket Bonus +10; 5-Wicket Bonus +15; Maiden Over +6.

Economy Rate: Minimum 2 overs bowled. <5.00: +5; 5.00–5.99: +3; 6.00–7.00: +1; 7.01–9.99: 0; 10.00–11.00: −1; 11.01–12.00: −3; >12.00: −5.

Fielding: Catch +6; 3-Catch Bonus +4; Stumping +10; Run Out (Direct Hit) +10; Run Out (Not a Direct Hit) +5.

Participation: Appearance in official Playing XI +3; players who do not play receive 0 points.

Only the highest applicable batting milestone and highest applicable wicket-haul bonus will be awarded. No points are awarded for events occurring during a Super Over. Official completed-match scorecard/ event data will be treated as authoritative.

Optional Bonus: Team Overviews via Local LLMs

You may optionally integrate a lightweight LLM to generate a brief text overview of your predicted team (e.g., explaining the composition or highlighting key player strengths).

You must make use of open-weight LLMs. API calls to commercial models are **strictly prohibited**.

Repository & Code Structure Requirements

- **Private Repository:** Your repo must be private. We will request collaborator access closer to the evaluation phase.
- **Modular Architecture:** Folder names are flexible, but concerns must be separated — raw data, processed data, model artifacts, training, and evaluation each in their own location.
- **Strict CLI Execution:** Do not hardcode paths or dates — use `argparse` or similar.
    - Training: `python src/train.py --train-start 2000-01-01 --train-end 2025-12-31`
    - Evaluation: `python src/evaluate.py --model-path model_artifacts/model_2025-12-31.pkl --test-file data/raw/test_fixtures.json --output results/hidden_predictions.csv`
- **Deterministic Outputs:** All random seeds must be fixed globally at the top of your scripts. Outputs must be 100% reproducible during the technical viva.
- **Reproducibility:** Include a root-level `run.sh` (or a comprehensive `Makefile`) that sequentially executes the full pipeline — data downloading, parameter checks, training, and final CSV generation.
- **Feature Explainability:** Interpret and discuss the results in your report — feature importance, the model's approximation behaviour, ablation studies, etc.
- **Leakage Audit:** Your report must include a short, explicit statement of how you verified no post-cutoff data reached training (e.g., unit tests on feature timestamps, manual spot-checks).

Mandatory Deliverables

- **GitHub Repository:** Clean, modular, reproducible code with a complete `README.md` covering setup, dependencies (`requirements.txt`), and run instructions.
- **Hidden Test Set Predictions:** A generated CSV containing your team predictions for the test file.
- **Presentation Deck (PPT):** High-level architectural walkthrough, design choices, feature engineering strategies, and result benchmarks.

- **Technical Report:** Methodology, ablations/experiments, quantitative metrics on your internal validation sets, model explainability, selection-algorithm justification, and the leakage audit above.

Evaluation

- Approach and architecture —---------- 20%
- Outputs, implementation, and experiments —- 25%
- Presentation, report, and demo —----------15%
- Interview —---------- 40%

Timeline

- **Hackathon Kickoff:** 22nd September
- **Hidden Test Data Release:** After 30th September
- **Submission Deadline:** 30th September, 11:59 PM
- **Interviews & Vivas:** 1st week of October

Notes

- Please refer to the live version of this document — it may be updated frequently.
- Novelty is always appreciated!
- If there are any doubts whatsoever, message in the *group*; we will respond immediately.